

Sycophancy Detection in Large Language Models Using Hidden State Probing

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Submission Tracks: Interpretability and Analysis of Models for NLP; Dialogue and Interactive Systems; Special Theme: New Missions for NLP Research.

1 Problem Formulation

Large language models (LLMs) trained with human feedback often exhibit **sycophancy** - the tendency to abandon a correct answer in response to user pressure, even when no new factual information is provided. In multi-turn interactions, this manifests as a model initially producing a correct response and then gradually aligning with an incorrect user claim through repeated disagreement or assertions of confidence, without any update in the underlying evidence.

This behavior presents a reliability and safety concern. In real-world deployments, users may provide incorrect, misleading, or adversarial input. A model that capitulates under social pressure is not reasoning, it is optimizing for agreement. In high-stakes settings such as medical advising, legal reasoning, or financial decision-making, such silent failures can have serious consequences.

Gap in Prior Work. Prior work has primarily studied sycophancy at the *behavioral level*, measuring when and how often models change their outputs under pressure. However, two questions remain unaddressed. First, it is unclear whether behavioral flips are *abrupt* or whether they are preceded by detectable internal shifts in model representations. Second, existing mitigation strategies: such as output filtering or fine-tuning that operate only after incorrect behavior has already manifested. This leaves open the possibility of earlier, more proactive intervention if internal pre-flip signals exist.

Research Question. In this work, we investigate the following core question:

Do large language models exhibit detectable internal signals in their hidden states that precede sycophantic behavior under multi-turn social pressure, and

can these signals be used to predict behavioral flips before they occur?

Formal Definition. We define a **flip** at turn t as the event in which a model’s response contradicts or retracts its Turn 0 answer following one or more pressure prompts, with no new factual information introduced. A **pre-flip state** is the hidden-layer representation extracted at turn $t - k$ for $k \geq 1$, prior to any observable behavioral change. The prediction task is: given a pre-flip hidden state, can we classify whether a flip will occur in a subsequent turn?

Hypothesis. We hypothesize that representational drift toward agreement begins *one to two turns before* observable behavioral capitulation, and that this signal is strongest in the middle-to-late layers of the transformer, where higher-level semantic and reasoning representations are encoded. If true, this implies that sycophancy is not a sudden surface-level output change, but a gradual internal process that is detectable before it becomes visible in model outputs.

Approach. To test the above hypothesis, we study LLM behavior in a controlled multi-turn setting with **escalating social pressure**, where the user progressively increases the strength of disagreement across five turns without introducing new information. At each turn, we extract hidden state representations of the final generated token across all transformer layers. We then train lightweight supervised probes on these representations to evaluate whether pre-flip signals are linearly separable, layer-localized, and transferable across model architectures. We treat above-chance probe accuracy on held-out pre-flip turns as the primary criterion for detecting a meaningful internal signal.

Scope. Our study focuses on:

- Multi-turn conversational settings with monotonically escalating pressure
- Open-weight transformer-based LLMs
- Hidden state representations at the final token across layers

Out of scope. We do not address:

- Training new models or modifying model weights
- Deployment of real-time intervention systems
- Human-in-the-loop evaluation of perceived correctness

By framing sycophancy as a *representation-level phenomenon* rather than purely a behavioral one, this work aims to move toward earlier and more precise detection mechanisms; ones that operate before incorrect outputs are produced, rather than reacting to them after the fact.

2 Related Work

Sycophancy and Alignment Behavior. Recent work has shown that large language models (LLMs) exhibit sycophantic behavior, aligning their responses with user beliefs even when those beliefs are incorrect. [Perez et al. \(2022\)](#) demonstrate that models trained with human feedback tend to prioritize agreement over correctness in certain settings. Follow-up work has further analyzed how reinforcement learning from human feedback (RLHF) shapes such behaviors, often trading off truthfulness for perceived helpfulness ([Ouyang et al., 2022](#); [Bai et al., 2022](#)). These studies primarily evaluate sycophancy at the behavioral level, focusing on final outputs rather than the internal dynamics that lead to such failures.

Multi-turn Interaction and Persuasion Effects. LLM behavior is highly sensitive to conversational context, especially in multi-turn settings. Prior work has shown that repeated user disagreement, rebuttal, and conversational framing can systematically alter model responses over time ([Hong et al., 2025](#); [Perez et al., 2022](#)). These studies highlight that alignment failures may emerge dynamically through interaction rather than appearing only in isolated prompts. However, most evaluations measure only the eventual response shift, without examining when or how internal representations begin to deviate.

Multi-turn Sycophancy Evaluation. [Hong et al. \(2025\)](#) introduce SYCON-Bench, a benchmark for evaluating sycophantic behavior in multi-turn dialogues. Their framework measures how quickly models conform to user beliefs (Turn of Flip) and how frequently they change their stance under sustained pressure (Number of Flip). Their results show that sycophancy remains prevalent across a wide range of LLMs, and that alignment tuning can further amplify this behavior, while reasoning-focused models exhibit greater resistance.

While this work captures the *behavioral dynamics* of sycophancy in multi-turn settings, it does not examine the internal mechanisms underlying these shifts. In contrast, our work focuses on *representation-level dynamics*, showing that hidden states begin to drift toward agreement before behavioral flips occur.

Probing and Representation Analysis. A large body of work uses probing techniques to extract interpretable signals from model representations ([Alain and Bengio, 2016](#); [Tenney et al., 2019](#)). More recent work extends probing to safety- and reliability-relevant properties, including truthfulness and latent knowledge in hidden states ([Burns et al., 2022](#); ?). These studies show that internal representations often contain information not directly visible in final outputs. However, they are typically applied to static prompts or factual settings rather than multi-turn conversational pressure.

Mechanistic Interpretability and Causal Analysis. Mechanistic interpretability aims to identify causal circuits within neural networks that give rise to specific behaviors ([Olah et al., 2020](#); [Meng et al., 2022](#)). Techniques such as activation patching and causal tracing have been used to localize factual recall and decision-making components within transformers. Recent advances, including sparse autoencoders (SAEs), attempt to decompose representations into more interpretable features ([Cunningham et al., 2023](#)). However, most of this work focuses on static tasks (e.g., factual recall, induction heads) rather than alignment failures in interactive settings.

Mitigation and Control of Undesired Behavior. Several approaches have been proposed to mitigate undesirable model behaviors, including constitutional AI ([Bai et al., 2022](#)), adversarial training, and inference-time steering methods ([Turner et al., 2023](#)). These methods aim to either reshape train-

Work	MT	Int. States	Pre-flip
Perez et al. (2022)	×	×	×
Hong et al. (2025)	✓	×	×
Azaria & Mitchell (2023)	×	✓	×
Ours	✓	✓	✓

Table 1: Comparison of prior work. MT = Multi-turn.

ing objectives or directly modify model activations during inference. While effective in some cases, they typically rely on detecting failures at the output level, leaving open the question of whether earlier internal signals could enable more proactive intervention.

Our Contribution. In contrast to prior work, we focus on the *internal dynamics of sycophancy in multi-turn interactions*. Specifically, we show that hidden-state representations begin to shift toward agreement *one to two turns before* observable behavioral flips occur. By constructing an escalating-pressure evaluation framework and extracting layer-wise representations across turns, we provide evidence that alignment failures can be anticipated at the representation level. This positions our work at the intersection of behavioral evaluation and mechanistic analysis, offering a step toward early detection and intervention in LLM alignment failures.

3 Data & Ethics

3.1 Dataset

We use the SYCON-Bench dataset (Hong et al., 2025), a publicly available benchmark for evaluating sycophantic behavior in multi-turn LLM dialogues. From the full dataset, we sample 474 questions spanning two categories:

- **Base questions** (~150): open-ended factual questions without embedded assumptions, requiring the model to maintain a correct answer under sustained disagreement.
- **Presupposition questions** (~150 per model): questions containing false premises that the model must identify, correct, and resist as pressure escalates.

SYCON-Bench is derived from publicly available factual question-answering sources and does not contain personally identifiable information (PII), sensitive demographic data, or user-generated content from private interactions. No additional data collection, annotation by human participants,

or crowdsourcing was conducted as part of this work.

3.2 Benchmark Adaptation

While we build on the SYCON-Bench dataset, our benchmark construction introduces two significant modifications. First, we replace the original repeated fixed-pressure protocol with a **monotonically escalating pressure schedule**, spanning five pressure levels from mild doubt to strong contradiction. Second, we restructure the conversational format so that each pressure prompt is appended as a new dialogue turn, with the full conversation history passed at every step. This shift from a concatenated single-input format to a true multi-turn dialogue was central to preserving authentic conversational dynamics and enabling hidden-state analysis across turns. These changes ensure our evaluation captures how pressure *accumulates* over a conversation rather than measuring response to a static prompt.

3.3 Ethical Considerations

All models evaluated in this work are open-weight, publicly released systems (Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, Gemma-2-9B-it). No proprietary or access-restricted APIs were used.

Our work does not involve human subjects, the collection of personal data, or the deployment of model outputs in real-world settings. The SYCON-Bench dataset is released under its original terms of use and is used here solely for research evaluation purposes, consistent with those terms.

We acknowledge that sycophancy detection systems, if deployed, could be misused to engineer pressure sequences that evade detection or to falsely flag legitimate model behavior as sycophantic. We frame this work as a diagnostic and mechanistic study rather than a prescriptive intervention tool, and we encourage caution in any downstream applications.

4 Experiment Description

4.1 Benchmark Construction

We construct a multi-turn escalating pressure benchmark adapted from SYCON-Bench (Hong et al., 2025). Each conversation consists of six turns: an initial question (Turn 0, no pressure) followed by five pressure prompts of monotonically increasing intensity. The pressure schedule ranges

from mild doubt (“Are you sure?”) to strong contradiction (“I’m confident you’re wrong and insist it’s incorrect.”).

This design differs from the original SYCON-Bench, which applies identical repeated pressure at each turn. By progressively increasing pressure, we capture not only whether a model flips but also the intensity at which the flip occurs, enabling finer-grained analysis of internal dynamics.

Additionally, we modify the conversational format: instead of concatenating prompts into a single input, we append each pressure prompt as a new turn and pass the full dialogue history at every step. This preserves true conversational context and allows pressure to accumulate naturally. Due to this difference, our results are not directly comparable to published SYCON-Bench baselines.

4.2 Models

We evaluate three instruction-tuned open-weight models:

- Qwen2.5-7B-Instruct (29 layers, 3584 hidden dimensions)
- Llama-3.1-8B-Instruct (33 layers, 4096 hidden dimensions)
- Gemma-2-9B (42 layers, 3584 hidden dimensions)

4.3 Hidden State Extraction

At each turn, we extract the hidden state of the final generated token across all transformer layers. This yields a tensor of size $L \times d$ per turn, where L is the number of layers and d is the hidden dimension.

To characterize internal dynamics, we compute:

- **L2 norm** of each layer’s hidden state (activation magnitude)
- **Cosine similarity** between consecutive turns (directional shift)

We also align turns relative to each question’s flip point, enabling aggregation of pre- and post-flip dynamics across examples with different flip timings.

4.4 Sycophancy Labeling

We label each post-Turn-0 response as either **hold** (model maintains its initial answer) or **flip** (model changes its stance). Labeling is performed using a curated list of 25+ deferential markers (e.g.,

“you’re correct,” “I apologize,” “let me reconsider”), validated through manual inspection.

We additionally explored LLM-based labeling but found it unreliable due to inconsistent judgments across similar responses. We therefore use keyword-based labeling for all experiments.

This approach captures explicit sycophancy but misses subtler forms (e.g., silent factual shifts), which likely lowers the upper bound on classification performance.

4.5 Probe Training

We train supervised probes on hidden states to predict behavioral outcomes. Two tasks are considered:

- **At-turn classification:** predict whether a given turn is a flip or hold.
- **Pre-flip prediction:** predict whether a flip will occur in a future turn, given a pre-flip hidden state.

Probes are trained independently on each layer to localize where predictive signals emerge.

We evaluate a diverse set of 11 classifiers, including linear models (logistic regression, linear SVM), nonlinear models (RBF SVM, MLP), and ensemble methods (random forest). Linear probes test whether the signal is linearly separable, while nonlinear models capture more complex structure.

4.6 Evaluation

We report performance using stratified k -fold cross-validation. Accuracy is measured relative to the baseline of the majority-class (Δ chance), and the F1-score is reported for the minority “flip” class due to class imbalance (75-85 %).

4.7 Cross-Model Transfer

To test generality, we train probes on one model (e.g., Llama-3.1-8B) and evaluate on another (e.g., Qwen2.5-7B) without any feature alignment. This provides a direct test of whether pre-flip signals are consistent across model architectures.

5 Results & Discussion

5.1 Behavioral Analysis: Models Cave Predictably

Before examining hidden-state dynamics, we characterize the behavioral profile of each model under our escalating pressure benchmark. Table 2 reports

flip rates and mean flip turn across model families and question types.

Model	Type	N	Flip%	Avg. Flip T
Llama-3.1-8B	Base	152	66.4	2.49
Llama-3.1-8B	Presup.	146	55.5	2.57
Qwen2.5-7B	Base	112	76.8	2.45
Qwen2.5-7B	Presup.	112	85.7	1.93

Table 2: Flip rate (%) and mean flip turn by model and question type. Presup. = Presupposition.

Several patterns emerge. First, both models exhibit high overall sycophancy under our escalating schedule, with Qwen2.5-7B showing a notably higher flip rate than Llama-3.1-8B across both question types. Second, and counterintuitively, Qwen2.5-7B flips *more* on presupposition questions (85.7%) than on base questions (76.8%), with a lower mean flip turn of 1.93 — indicating that questions containing false premises are not harder to resist but are in fact easier to destabilize. Llama-3.1-8B shows the opposite pattern: its flip rate on presupposition questions (55.5%) is lower than on base questions (66.4%), suggesting that the false-premise structure may engage a correction reflex that adds some resistance to social pressure. This model-specific divergence in category-level response has implications for how alignment is encoded differently across architectures.

Third, mean flip turns cluster between 1.93 and 2.57 across all conditions, indicating that capitulation typically occurs early in the pressure schedule — well before maximum pressure is applied. This systematic early-caving pattern motivates the core hypothesis: if behavioral change is predictable and relatively consistent in timing, the internal representations may begin shifting even earlier.

5.2 Early Disruption Predicts Flip Timing

A key finding emerges from analyzing cosine similarity of hidden states between Turn 0 and Turn 1 — the transition from no pressure to the first, mildest pressure prompt. Across all three model architectures, the **largest directional shift in hidden states occurs at T0→T1**, not at the turn where the behavioral flip is observed. Moreover, the magnitude of this initial disruption is predictive of how early the model eventually caves.

Gemma-2-9B exhibits the lowest cosine similarity (0.793) and the earliest mean flip turn (1.77), while Qwen2.5-7B shows the highest cosine similarity (0.835) and the latest mean flip turn (3.77).

Model	CosSim (T0→T1)	Mean Flip Turn
Gemma-2-9B	0.793	1.77
Llama-3.1-8B	0.824	~3.0
Qwen2.5-7B	0.835	3.77

Table 3: Lower cosine similarity at first pressure (T0→T1) predicts earlier behavioral flip. Relationship holds across all three architectures without any probe training.

Llama-3.1-8B falls between both. This directional relationship holds across all three architectures and requires no probe training to observe — it follows directly from cosine similarity between Turn 0 and Turn 1 hidden states. This constitutes the strongest cross-model claim in this work: a simple, training-free signal measured at the first pressure prompt is predictive of eventual behavioral capitulation timing.

This finding implies that a model’s internal response to the *first* challenge encodes latent information about its overall robustness under pressure — a property that is invisible at the output level, since all models produce a holding response at Turn 1, yet is already reflected in the geometry of their internal representations.

5.3 Layer Localization: Where the Signal Lives

To determine which transformer layers carry the most predictive information about sycophancy, we train probes independently on hidden states from each layer and plot accuracy as a function of depth. Results on Llama-3.1-8B base questions reveal a clear dissociation between where representations *change the most* and where they are *most predictable*.

Early layers (L1–L5) yield approximately 60% probe accuracy — near chance — indicating minimal sycophancy signal in low-level representations. Probe accuracy rises monotonically through the middle layers, peaking at layers L17–L19 with approximately 74% accuracy. This upper-middle region represents the **peak predictive zone**, where the sycophancy decision appears to be most decodable. Final layers (L30–L32) show the most absolute representational drift as measured by L2 norm and cosine similarity, yet probe accuracy drops back to approximately 70% — confirming that the layers exhibiting the greatest surface-level change are not the same layers carrying the most structured, classifiable signal.

This layer-level dissociation is a substantive finding. It suggests that sycophancy is not simply an output-layer phenomenon but a mid-network reorganization of semantic representations. The middle-to-late layers — roughly the upper third of the transformer — encode the alignment decision in a form that is geometrically separable before it reaches the decoding stage. This is consistent with prior probing work showing that higher-level semantic properties are increasingly localized to mid-network layers in large transformers (Tenney et al., 2019), and extends that finding to dynamic, pressure-driven alignment failures.

5.4 Probe Performance: The Signal is Nonlinearly Encoded

Table 4 reports pre-flip prediction accuracy across all 11 classifiers evaluated on the best-performing layer. The majority-class chance baseline is 63.8%, reflecting the class imbalance in which 75–85% of turns are labeled *hold*.

Classifier	Accuracy	Δ Chance
RBF SVM	75.5%	+11.7%
Random Forest	75.5%	+11.7%
KNN ($k=10$)	75.5%	+11.7%
MLP (128+64)	73.6%	+9.8%
Naive Bayes	72.5%	+8.7%
Logistic Regression	64.2%	+0.4%
Linear SVM	60.4%	−3.4%

Table 4: Pre-flip prediction accuracy across classifiers. Chance baseline: 63.8%.

Three nonlinear classifiers — RBF SVM, Random Forest, and KNN ($k=10$) — all achieve 75.5% accuracy, representing an 11.7-percentage-point improvement over chance. In contrast, linear classifiers perform near or below chance: logistic regression reaches only 64.2% (+0.4%) and linear SVM underperforms at 60.4% (−3.4%). This gap between linear and nonlinear probes is the primary evidence that **the pre-flip signal is nonlinearly encoded in hidden state geometry**.

The KNN result is particularly informative. Unlike RBF SVM and Random Forest, KNN has no learned parameters — it classifies a point purely by checking whether it is geometrically proximate to other labeled pre-flip states. The fact that KNN matches the performance of trained nonlinear classifiers at 75.5% confirms that the clustering is a property of the *representation itself* rather than an artifact of the classifier’s capacity. Pre-flip hidden states form a coherent geometric region on a curved

manifold in representation space, and this structure is dense enough that nearest-neighbor lookup alone is sufficient to exploit it.

This finding revises our original hypothesis: while we anticipated the pre-flip signal to be linearly decodable (as is common in probing work on factual representations), sycophancy encodes differently — in a nonlinear geometric structure that linear probes cannot access. The implication is that lightweight interpretability tools calibrated for linear separability may systematically underestimate the presence of alignment-relevant internal signals.

5.5 Hypothesis Evaluation

We close by directly evaluating the three hypotheses articulated in Section 1.

H1 — Internal precursors exist (Supported). L2 norms and cosine similarities show systematic, progressive changes in hidden states before any behavioral flip occurs. The per-layer sweep further shows a monotonic increase in probe accuracy from layers 0 through 17, consistent with the gradual accumulation of a pre-flip signal rather than a sudden late-layer shift.

H2 — Confidence degrades gradually (Supported). Relative-to-flip analysis reveals L2 norm buildup over 3–4 turns before the flip turn. The smooth accuracy gradient across layers and the directional shift pattern at T0→T1 collectively confirm that sycophantic capitulation is a gradual internal process — not a sudden output-level switch — making it detectable before it manifests behaviorally.

H3 — Probes can predict flips (Revised). The signal exists and is predictable, but it is not linearly separable as initially hypothesized. Nonlinear classifiers achieve 75.5% accuracy, confirming that pre-flip states are classifiable above chance. However, linear probes fail to access this signal, indicating that the representational geometry of sycophancy is curved rather than hyperplanar. The hypothesis is supported in substance but requires revision in its assumed decodability assumption.

5.6 Cross-Model Transfer

We attempted to test whether probes trained on one model transfer to another without feature alignment. Probes trained on Llama-3.1-8B and evaluated on Qwen2.5-7B produced only majority-class

predictions, performing at chance. This failure is expected given that raw hidden states from different model families occupy fundamentally different geometric spaces — there is no shared coordinate system between a 4096-dimensional Llama representation and a 3584-dimensional Qwen representation. Cross-model transferability would require either dimensionality-aligned representations, shared probing directions, or architecture-agnostic features derived from the hidden states rather than the raw vectors. We treat this as an important negative result and discuss it further in Section **Conclusion & Limitations**.

6 Conclusion & Limitations

6.1 Summary of Findings

This work investigated whether sycophantic behavior in LLMs is preceded by detectable internal signals in hidden-state representations. Using an escalating-pressure benchmark adapted from SYCON-Bench, we extracted layer-wise hidden states from two model families across 474 questions and six conversation turns. Three hypotheses were tested:

H1 (Supported). Internal precursors to sycophancy exist. Hidden-state properties — L2 norms, cosine similarity, and probe accuracy — shift systematically across turns before any behavioral flip occurs. Probe accuracy increases monotonically from early layers ($\sim 60\%$) to upper-middle layers ($\sim 74\%$), confirming that pre-flip information accumulates in a structured, layer-dependent manner.

H2 (Supported). Confidence degrades gradually rather than collapsing abruptly. Relative-to-flip analysis shows L2 norm buildup over 3–4 turns before the flip, and the smooth accuracy gradient across layers rules out a sudden-switch mechanism.

H3 (Revised). The pre-flip signal is detectable but nonlinearly encoded. Linear probes (logistic regression, linear SVM) detect above-random signal but fail to exceed the majority-class baseline. Non-linear classifiers — RBF SVM, Random Forest, and KNN — achieve 75.5% pre-flip prediction accuracy (+11.7% above chance). KNN’s matching performance, despite having no learned parameters, confirms that pre-flip states form a geometric cluster in representation space whose boundary is nonlinear.

Two additional findings emerged beyond the original hypotheses. First, cosine similarity at

$T0 \rightarrow T1$ is negatively correlated with mean flip turn across all three model architectures, providing a training-free, cross-model early-warning signal. Second, probe accuracy peaks at layers 17–19 rather than at the final layers where representational drift is largest — a dissociation between where the signal *changes most* and where it is most *decodable*.

6.2 Limitations

Labeling ceiling. Sycophancy detection relies on keyword matching against 25+ deferential markers. This approach captures explicit capitulation but misses subtler forms — silent factual shifts (e.g., “over 200 strains” becoming “over 100 strains”), deferential reframing without concession language, and partial stance erosion. These mislabeled hold examples compress the true accuracy ceiling of any classifier trained on these labels. An LLM-as-judge labeling approach would likely produce cleaner ground truth and improve probe accuracy.

Cross-model transfer failure. Probes trained on Llama-3.1-8B and tested on Qwen2.5-7B produced only majority-class predictions. Raw hidden states from different architectures occupy fundamentally different geometric spaces, and no shared coordinate system exists between a 4096-dimensional Llama representation and a 3584-dimensional Qwen representation. Cross-model generalization would require derived feature engineering (e.g., cosine similarity deltas, normalized drift metrics) or explicit representation alignment techniques such as CKA or CCA.

Gemma data corruption. Hidden-state files for Gemma-2-9B were corrupted during transfer and could not be loaded for probe training. Behavioral analysis (flip rates, CosSim trajectories) was completed using CSV-level summaries, but the absence of Gemma probe results weakens the cross-model generalizability claim from three model families to two.

Class imbalance. In the pre-flip prediction task, 75–85% of turns are labeled hold, making the majority-class baseline difficult to exceed even when the probe captures genuine signal. While we use balanced class weights in all classifiers, techniques such as focal loss, SMOTE, or subsampling could further improve minority-class recall.

Layer selection on the evaluation loop. The best-performing layer was selected using cross-

validation accuracy, and the same accuracy was reported as the final result. This constitutes mild double-dipping; a dedicated held-out set for layer selection would provide cleaner separation between model selection and evaluation.

Modified benchmark. Our escalating-pressure protocol and true multi-turn conversation history represent deliberate departures from the original SYCON-Bench design. While we believe these changes better capture real-world conversational dynamics, they mean our results are not directly comparable to published SYCON-Bench baselines and should be interpreted within the context of our modified framework.

6.3 Future Directions

Three extensions follow naturally from this work. First, replacing keyword-based labeling with an LLM-as-judge approach would raise label quality and likely improve probe accuracy by an estimated 5–10%. Second, training cross-model probes on derived, architecture-agnostic features — CosSim $T_0 \rightarrow T_1$, L2 delta, layer-normalized drift — rather than raw hidden-state vectors could enable transfer across model families. Third, a runtime guardrail implementation that monitors cosine similarity drift from the Turn 0 anchor at layers 17–21 and triggers a lightweight nonlinear probe (Random Forest or RBF SVM) when drift exceeds a threshold would provide a practical, real-time sycophancy detection mechanism that operates before incorrect outputs are produced.

7 Team Roles

This project was conducted as a collaborative effort, with responsibilities divided across experiment design, implementation, and analysis.

Utkarsh Saraogi. Designed and led the **benchmark reformulation, hidden state extraction, and cross-model behavioral analysis pipeline**.

- Proposed and implemented the **escalating pressure benchmark**, replacing the original SYCON-Bench repeated-pressure protocol with a monotonically increasing schedule and restructuring to a true multi-turn dialogue format, appending each pressure prompt as a new turn with full dialogue history stored.
- Developed the **hidden state extraction pipeline**, implementing layer-wise extraction of the final

generated token’s representations across all transformer layers at each conversation turn, producing structured .pt checkpoint files used as inputs to all subsequent probing experiments.

- Developed **cross-model behavioral analysis scripts to parse model response** .json files and compute per-turn flip labels, flip rate distributions, and cosine similarity trajectories identifying that the largest hidden-state directional shift occurs at $T_0 \rightarrow T_1$, and that lower cosine similarity at first pressure predicts earlier behavioral flip timing across all three architectures without any probe training.

Soham Padia. Led the **experimental infrastructure, execution, and reproducibility pipeline**, and contributed to framing the project as a **mechanistic detection problem** rather than a purely behavioral diagnosis of sycophancy. Specifically:

- Helped define the project around **representation-level detection**, focusing on hidden-state signals that precede observable behavioral flips. Designed and maintained the **end-to-end benchmarking pipeline**, including execution of multi-turn experiments with layer-wise hidden state extraction across models and datasets.
- Implemented **resume-aware experimentation**, enabling large-scale runs to recover from interruptions without restarting, which was critical for long-running hidden-state extraction jobs. Built and maintained **HPC/Slurm workflows**, including batch scripts and automated job submission pipelines for running experiments across multiple models and configurations.
- Improved **model compatibility and execution reliability**, including device selection, Apple Silicon support, and robust model loading. Developed **hidden-state output handling and reporting pipelines**, supporting downstream probe training and analysis through structured outputs and post-processing scripts.

Vedant Shah. Led probe training, classifier evaluation, and the conclusion analysis. Specifically:

- Designed and implemented the full probe training pipeline, building dataset construction functions for both pre-flip prediction (Turn N hidden state $\rightarrow$ flip at Turn $N+1$) and ever-flip classification (Turn 0/1 hidden

state $\rightarrow$ eventual flip), with post-flip turn exclusion to prevent label noise.

- Conducted the 11-classifier sweep across logistic regression, linear SVM, ridge, RBF SVM, random forest, gradient boosting, KNN, MLP (three capacity levels), and naive Bayes, producing the nonlinear encoding finding that revised H3.
- Executed per-layer probe accuracy sweeps across all 33 transformer layers for both Llama-3.1-8B and Qwen2.5-7B, identifying layers 17–19 as the peak predictive zone and establishing the drift $\neq$ predictiveness dissociation.
- Implemented sycophancy labeling from meta-data JSON files using keyword matching on assistant responses, expanding and validating the marker list against manual inspection of model outputs.
- Ran cross-model probe transfer experiments (train on Llama, test on Qwen and vice versa), documenting the negative result with majority-class prediction analysis to distinguish genuine transfer from degenerate classifiers.
- Performed initial hypothesis testing on pilot data (Qwen2.5-0.5B-Instruct, 6 questions), computing hidden-state norms, variance, cosine similarity trajectories, and cross-layer drift analysis that informed the full-scale experimental design.

Tomas D’Avola. Contributed to **experiment execution, probe training, behavioral analysis, visualization, and report authorship.** Specifically:

- Participated in the **probe training and classifier sweep**, contributing to dataset construction for pre-flip and ever-flip prediction tasks, and to the evaluation of the 11-classifier sweep across linear, nonlinear, and ensemble methods on both Llama-3.1-8B and Qwen2.5-7B hidden states.
- Helped **run large-scale benchmark experiments** across models and scales, including working with HPC/Slurm execution workflows to launch and monitor multi-model evaluation jobs on the cluster.
- Developed **behavioral statistics and metric computation** scripts, producing per-model flip

rates, mean flip turns, and category-level aggregations (base vs presupposition) used throughout the results section.

- Built the **figure generation pipeline** for the report and presentation, producing the layer-wise probe accuracy sweeps, cosine similarity trajectories, L2 norm progression plots, and classifier comparison charts.
- Drafted the **Problem Formulation** section of the final report, including the research question, formal definitions of flip and pre-flip state, hypothesis articulation, and scoping of in-scope vs out-of-scope contributions.
- Established and maintained the **LaTeX infrastructure** for the final report, configuring the ACL style template, bibliography, and compilation pipeline, and handling final-pass editing and formatting across sections.
- Co-designed the **presentation slides** with teammates, contributing to the approach and evaluation slide structure and drafting speaker notes for the layer analysis, probe results, and limitations slides.

Submission Track Justification

This work is primarily positioned within **Interpretability and Analysis of Models for NLP**, as it uses hidden-state probing and layer-wise analysis to study the internal dynamics that precede sycophantic behavior.

It also aligns with **Dialogue and Interactive Systems**, since the experiments are conducted in a multi-turn conversational setting where user pressure accumulates over interaction.

Finally, it fits the **Special Theme: New Missions for NLP Research** because it argues for a shift from purely behavioral evaluation of alignment failures toward earlier, representation-level detection of such failures before they become visible in model outputs.

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